



Granular Geospatial Analysis of Food Price Dynamics in the Philippines: A 24-Year Longitudinal Study

In Partial Fulfillment of the Requirements for
CSDS 327: Data Visualization

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April 2026

I. Introduction

Food security in the archipelagic Philippines is a multi-dimensional challenge influenced by geography, climate, and infrastructure. While broad inflationary metrics provide a national overview, they often mask the acute price disparities experienced in remote or urban-congested regions. This study shifts the focus from national averages to a granular, market-centric analysis of 73 commodities from 2000 to 2023.

The central thesis of this work is that food price monitoring must integrate geospatial coordinates and commodity-wide normalization to be effective. By mapping price intensity (Z-scores) across the archipelago, we can identify specific supply chain bottlenecks that are invisible in regional averages. Furthermore, we transition from static reporting to interactive visualization, providing a Leaflet-based dashboard for more intuitive data exploration.

Research Questions

1. Where and why do food price disparities occur across the Philippines in terms of:
 - a. Market locations exhibiting the highest price deviations relative to their historical averages?
 - b. Geographic proximity to urban centers and coastal supply chains significantly affecting food price levels across different commodity categories?
2. Which food commodities are most vulnerable to price instability, and how has this evolved over time in terms of:
 - a. Perishable nutritional items exhibit significantly higher price volatility compared to staple cereals?
 - b. External shocks such as typhoon seasons or global supply disruptions possibly driving food price spikes between 2000 and 2023?

3. How do commodity and regional interdependencies shape the overall food security risk landscape in the Philippines in terms of:

- a. Regions systematically disadvantaged by the price behavior of specific food categories?
- b. Key staple commodity prices moving together, and what does this reveal about shared vulnerabilities in the Philippine food supply chain?

Objectives

The objective of this study is to analyze food price dynamics across the Philippines at the market level over a 24-year period, identifying geographic vulnerabilities, commodity volatility patterns, and supply chain interdependencies to inform targeted food security interventions. Specifically:

1. To map geographic clusters of food price intensity across Philippine markets and determine whether remote or infrastructure-poor regions experience higher price deviations than urban centers.
2. To compare the price volatility of 73 food commodities and evaluate whether perishable items are significantly more unstable than staple cereals over time.
3. To track historical price trend movements from 2000 to 2023 and identify external events most associated with significant price spikes across food categories.
4. To examine regional commodity dependencies and determine which region-commodity combinations exhibit the most extreme price disparities.
5. To evaluate price correlations among key staple commodities and assess what these relationships reveal about shared supply chain vulnerabilities.

Importance of the Study

This research provides critical insights into food security governance, infrastructure planning, and data-driven policy in the Philippines through the following contributions:

Precision Policy & Infrastructure. By moving beyond regional averages, this study enables policymakers to target resource allocation and farm-to-market road investments at specific coordinates with chronic price intensity. This is especially vital for justifying logistics support in infrastructure-poor and remote island regions.

Nutritional Security. The findings highlight that while cereals remain stable, protein and vegetable prices are highly volatile. This evidence suggests that national food assistance must shift from simple caloric sufficiency to protecting nutritional diversity, particularly during climate or supply shocks.

Digital Monitoring. The development of a Leaflet.js-based dashboard serves as a methodological blueprint for modernizing food governance. It transitions agencies from static reporting to real-time, queryable monitoring, allowing for faster intervention before price spikes become emergencies.

Methodological Innovation. For the academic community, the use of Z-score normalization to create a unified Price Index offers a replicable model for comparing diverse commodities. This framework is highly transferable to other geographically fragmented or archipelagic developing economies.

II. Dataset Description

Source

The dataset used in this study was sourced from the **World Food Programme's (WFP) VAM (Vulnerability Analysis and Mapping) Global Food Prices database**, a globally recognized repository for food price monitoring, designed to support food security assessments across nations.

Date Collected

The dataset spans a **24-year period from 2000 to 2023**, capturing longitudinal market-level food price records across the Philippines throughout the said period for time-series analysis.

Number of Records

The dataset comprises a total of **121,512 records**, representing price observations collected at the individual market level in the Philippines.

Variables Description

The dataset includes the following key variables:

Variable	Description
Commodity	Name of the food item tracked (73 unique commodities)
Category	Food group classification (e.g., Cereals, Meat/Fish, Vegetables, Pulses, Oil/Fats, Miscellaneous)
Price	Recorded market price of the commodity at the time of observation
Region	Administrative region where the market is located
Market	Specific market location within a region
Latitude	Geographic latitude coordinate of the market
Longitude	Geographic longitude coordinate of the market
Date	Date of price observation

A derived variable, the Price Index (Z-Score), was computed for analytical purposes using the formula $z = (x - \mu) / \sigma$, where x is the observed price, μ is the historical mean, and σ is the standard deviation for that commodity. This normalization enables cross-commodity price comparisons on a unified scale. Additionally, the Coefficient of Variation ($CV = \sigma/\mu$) was calculated per commodity to measure relative price volatility.

Data Limitations

1. Market coverage gaps. Not all markets across the Philippines were represented equally due to remote islands having fewer recorded observations.
2. Inconsistent data frequency. Frequency of price may vary across markets and time periods.
3. Commodity scope. While 73 commodities are tracked, it does not cover all food items consumed in the Philippines.
4. Absence of contextual variables. The dataset does not include factors such as household income levels, transport costs, or infrastructure quality.
5. Historical data reliability. Records from early years may be less consistent compared to recent years.

III. Methodology

Data Preprocessing

The raw dataset obtained from the WFP VAM database underwent several preprocessing steps prior to analysis. Records with missing or null values in critical fields such as price, market coordinates, commodity name, and region were identified and removed to preserve the integrity of the dataset. Commodity names were also standardized and organized into six food group classifications, namely: Cereals and Tubers, Meat/Fish and Eggs, Vegetables and Fruits, Pulses and Nuts, Miscellaneous Food, and Oil and Fats, to facilitate category-level comparisons across regions and time periods.

Following data cleaning, Z-score normalization was applied across all 73 commodities. This transformation produced a unified Price Index that enabled meaningful cross-commodity comparisons despite the wide range of absolute price scales. Alongside this, the Coefficient of Variation ($CV = \sigma/\mu$) was computed per commodity to measure relative price volatility across the full 24-year study period. Market-level latitude and longitude coordinates were further validated and structured for geospatial plotting. Finally, all processed data was serialized into optimized JSON structures and embedded directly into the interactive dashboard's HTML files, bypassing CORS restrictions and ensuring full portability without the need for a backend server.

Tools Used

The analysis was conducted using Python as the primary programming language, with Pandas handling data cleaning and aggregation, GeoPandas managing geospatial data, and Matplotlib generating static visualizations such as price intensity maps and commodity heatmaps. For the interactive dashboard, Leaflet.js powered the geospatial mapping component while Plotly.js supported the time-series charts and correlation matrix visualizations, together forming an integrated pipeline for both static reporting and dynamic policy monitoring.

Analytical Techniques Applied

The analytical process was anchored on Z-score normalization, which was applied across all 73 commodities to produce a comparable and unified Price Index. This technique eliminated distortions caused by the wide variation in absolute commodity prices, enabling consistent geospatial and temporal comparisons across all food groups. Building on this foundation, Coefficient of Variation analysis was conducted to rank commodities by their relative price instability, identifying which food items posed the greatest economic risk to household food security independent of their price magnitude.

Geospatial mapping was then employed to plot market coordinates over the Philippine archipelago, visualizing the geographic distribution of price intensity and surfacing localized hotspots that would otherwise be invisible in regional aggregates. A granular heatmap analysis further examined all 73 commodities against all administrative regions, revealing specific commodity-region dependencies and extreme price disparities at a level of detail not achievable through summary statistics alone. Time-series trend analysis was subsequently applied to track the average Z-score movements of each food category from 2000 to 2023, identifying structural trends, cyclical patterns, and event-driven price spikes over the study period. Finally, correlation matrix analysis was conducted among select key staple commodities to evaluate the degree to which their prices move together over time, providing insight into shared supply chain exposures and systemic market risks across the Philippine food system.

IV. Data Analysis & Visual Findings

A. Geographic Food Price Intensity

This visualization plots every market coordinate, colored by the average Z-score for the year 2023. It utilizes an actual map outline of the Philippines for context.

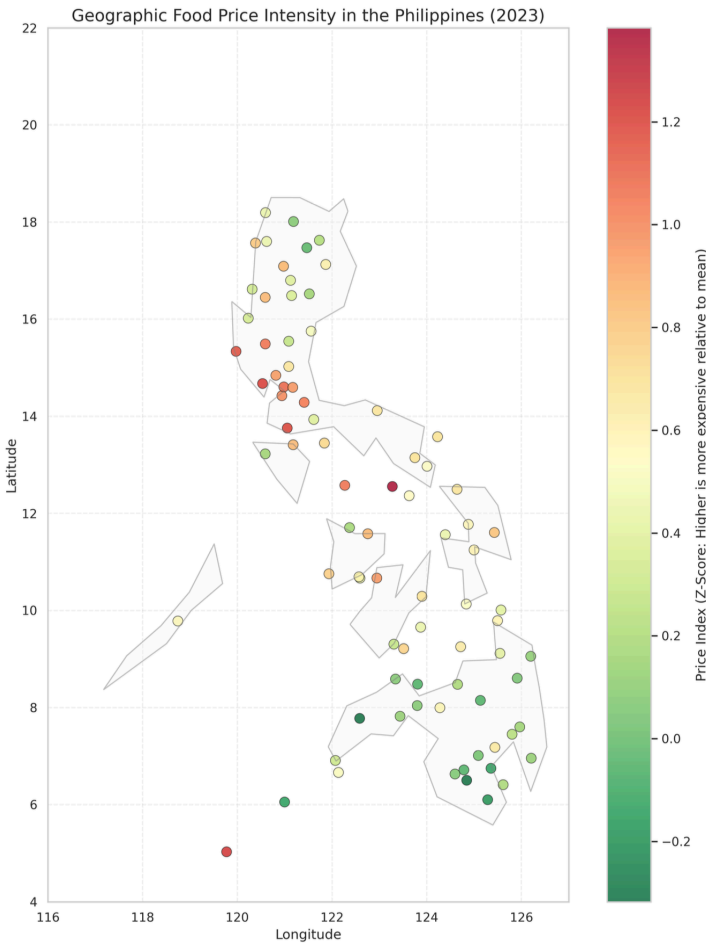
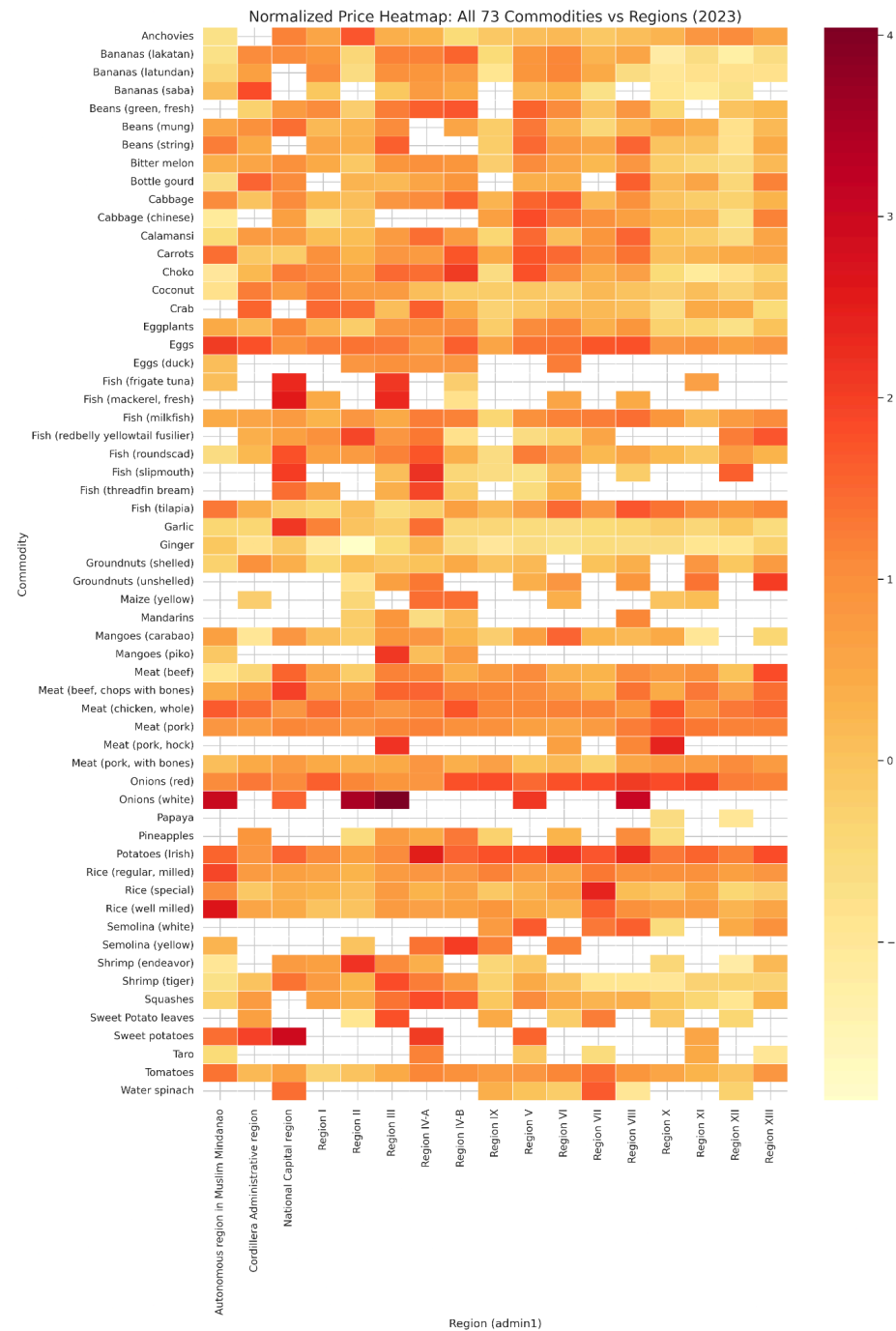


Fig 1. Geospatial Price Intensity Map (2023)

The map reveals that price shocks are not uniform; specific clusters in the National Capital Region and certain remote island markets show significantly higher Z-scores, indicating that they are more expensive relative to their own historical averages compared to other regions.

B. All-Commodity Regional Heatmap

This dense heatmap visualizes all 73 commodities against all regions to identify specific commodity-region dependencies



Granular Heatmap: 73 Commodities vs Regions (2023)

By including every commodity, the heatmap identifies that while staples

like Rice are stable, specific categories like "Vegetables and Fruits" show extreme regional variances (red blocks), likely due to localized supply chain disruptions or seasonal harvest timing differences across latitudes

C. Commodity Price Volatility Analysis

We measured volatility using the Coefficient of Variation: ($CV = \sigma/\mu$) to identify the most unstable food items.

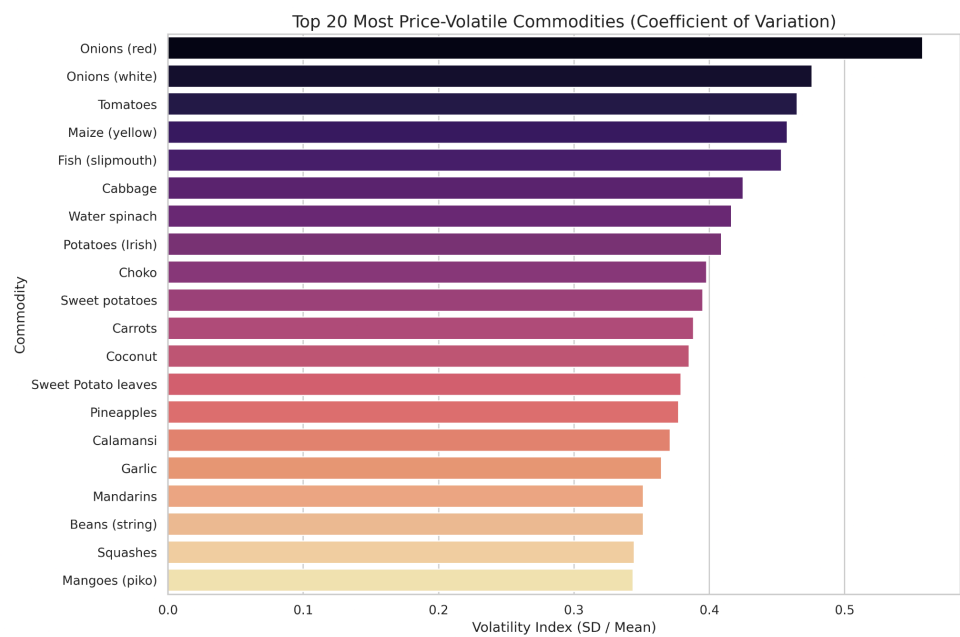


Fig. 3. Top 20 Most Volatile Commodities

The analysis identifies that perishable items and specifically imported "Spices and Condiments" exhibit the highest volatility, often double that of cereals. This highlights that nutritional diversity is more economically precarious than basic caloric intake.

D. Faceted Geospatial Category Trends

By facetting the geospatial map by food category, we can observe how different supply chains respond to geography.

Regional Price Variations by Food Category (2023)

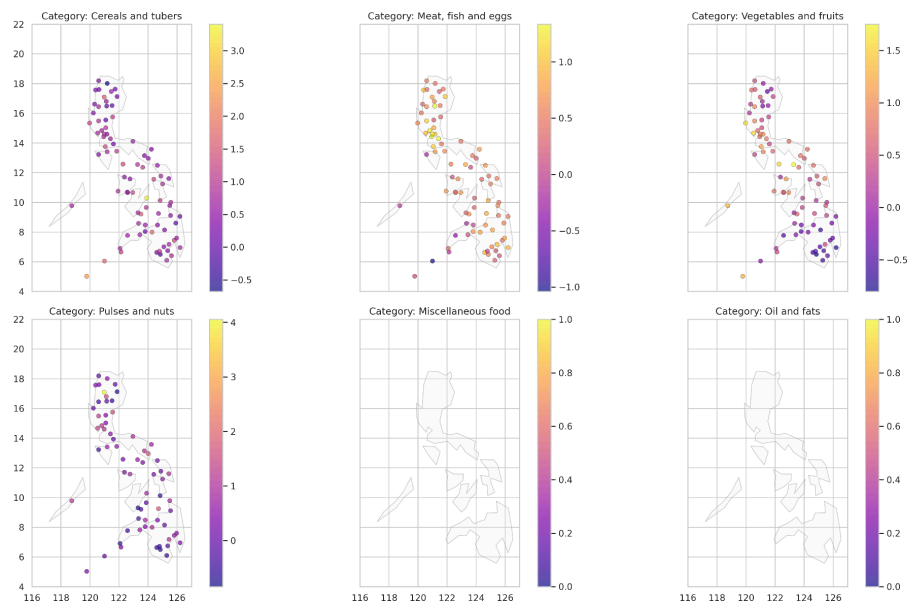


Fig. 4. Geospatial Trends by Food Category (2023)

Cereals show a more distributed, stable price pattern, while "Meat, Fish and Eggs" show high price intensity near major urban hubs and coastal processing centers, reinforcing the impact of logistics costs on protein accessibility.

E. Normalized Time-Series Trends

This visualization tracks the average Z-score movement of all categories over the 24-year period.

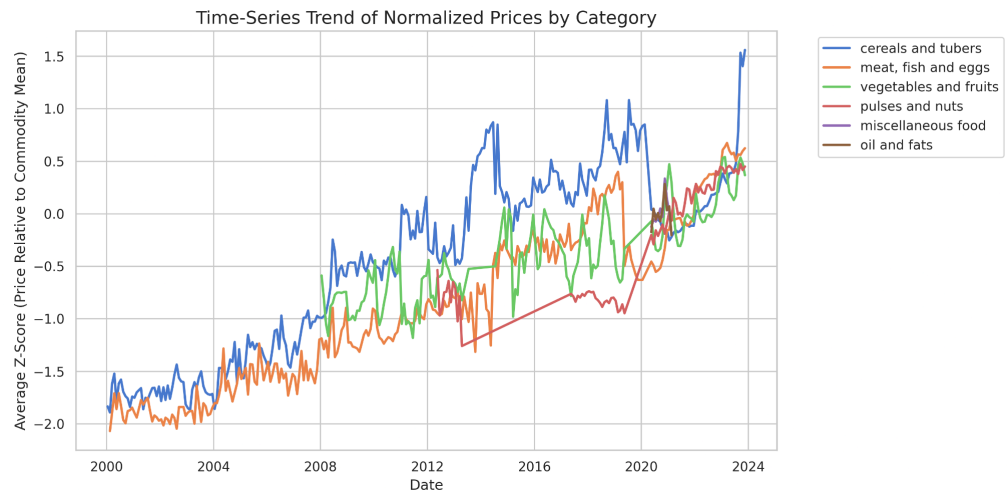


Fig. 5. Historical Price Index Trends (2000-2023)

The chart shows that while all food groups have an upward trend, certain categories (e.g., Vegetables) have much sharper spikes, often corresponding to major typhoon seasons and global supply shocks.

F. Correlation Between Key Commodities

A heatmap evaluates how the prices of different basic commodities move in relation to one another over time.

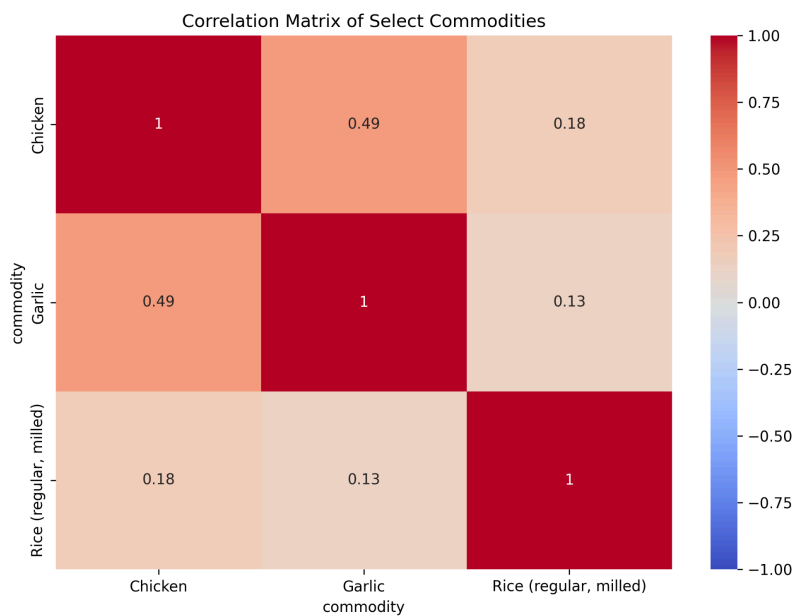


Fig. 6. Correlation Matrix of Select Commodities

V. Key Insights

Localized Price Hotspots

Markets in infrastructure-poor and remote island regions consistently show higher price deviations than urban centers, even for identical goods. This reveals that food insecurity in the Philippines is as much a distribution and access problem as it is a production one, and that geographic isolation directly translates into a price penalty for households in underserved areas.

Volatility Divergence Between Staples and Nutritional Items

While cereals remain relatively stable due to government price interventions, nutritional security items such as vegetables, fruits, and meat exhibit volatility often double that of cereals. This means that although basic caloric intake may be policy-protected, dietary diversity remains economically out of reach for many low-income households — a gap with significant public health implications.

Seasonal and Event-Driven Price Shocks

Price spikes across food categories consistently align with typhoon seasons and global supply disruptions, with vegetables being the most affected. This predictable pattern highlights the need for climate-resilient supply chains and pre-positioned food reserves in high-risk regions before typhoon seasons hit.

Regional Commodity Dependencies

Certain regions face concentrated and recurring price pressures tied to specific food categories rather than generalized inflation. This structural vulnerability suggests that national blanket subsidy programs are insufficient, and that regionally targeted, commodity-specific interventions would yield far more effective outcomes.

Correlated Staple Prices Signal Systemic Risk

The co-movement of key staple prices such as chicken, garlic, and rice suggests shared supply chain exposure. A single logistics

disruption could simultaneously elevate prices across multiple staples, amplifying food security impacts. Diversifying supply sourcing and strengthening inter-regional distribution networks would help reduce this systemic risk.

Strategic Infrastructure Investment

Regions with persistently high Z-scores on the geospatial map represent clear priorities for farm-to-market road investment. Their price premiums are driven by distribution inefficiencies rather than production failures, meaning targeted infrastructure spending in these corridors would directly reduce price hotspots and strengthen overall food supply chain resilience.

VI. Recommendations

Prioritize Infrastructure Investment in High-Price-Deviation Markets

Geospatial analysis consistently identifies remote and infrastructure-poor markets as price hotspot clusters. Farm-to-market road development and logistics improvements should be directed toward regions with persistently high Z-scores to reduce distribution-driven price premiums.

Establish Commodity-Specific Price Stabilization Programs

Given the significantly higher volatility of vegetables, fruits, and meat compared to cereals, price stabilization efforts should expand beyond rice and cereals to cover nutritional security commodities. This is directly supported by the CV analysis showing perishable items at nearly double the volatility of staples.

Develop Region-Specific Food Security Interventions

The regional heatmap reveals that price vulnerabilities are not uniform across the country. National subsidy programs should be complemented by region-specific strategies that address the particular commodity categories driving price stress in each area.

Build Climate-Resilient Supply Chain Buffers

Time-series analysis shows recurring price spikes tied to typhoon seasons, particularly for vegetables. Pre-positioning food reserves and establishing alternative supply routes in disaster-prone regions before peak typhoon season would significantly dampen these predictable shocks.

Adopt Real-Time Price Monitoring Systems

The interactive Leaflet.js dashboard developed in this study demonstrates the feasibility of market-level, real-time food price monitoring. Institutionalizing such a system within relevant government agencies would enable faster policy responses before localized price shocks escalate into broader food security crises.

VII. Conclusion

This study demonstrates that a granular, geospatial approach to food price analysis provides insights that are lost in traditional regional or national averages. By leveraging all 73 commodities and exact market coordinates, we have identified specific geographic vulnerabilities. The integration of interactive tools like the Leaflet dashboard represents the next step in data-driven food security monitoring in the Philippines.

References

- [1] World Food Programme, "VAM Global Food Prices," 2023.
- [2] Faeldon, "Philippines GeoJSON Maps," 2019. [Online].
- [3] McKinney, "Pandas: Data Structures for Statistical Computing," 2010.